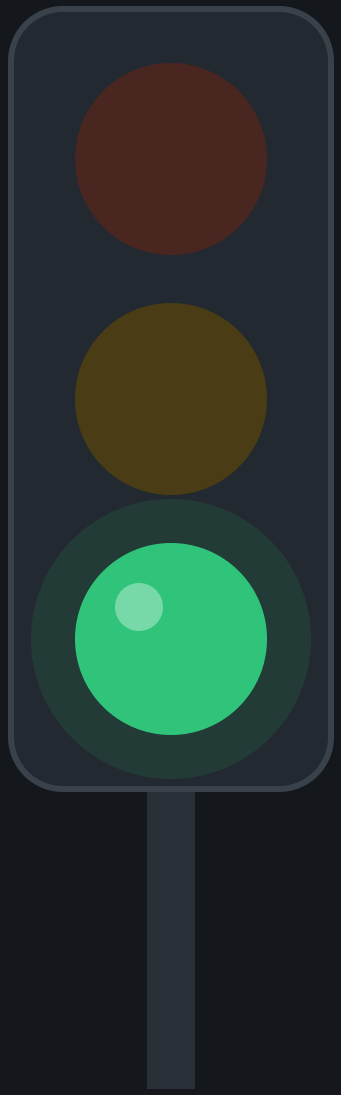
FOAI PBL PRESENTATION | REVIEW 1

Intelligent Traffic Signal Control Using Fuzzy Logic and Optimization

A signal that looks at the queue before it decides how long to stay green.



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Foundations of AI | Individual project

00 ROADMAP

What I will cover in the next few minutes

The talk follows the Review 1 evaluation points in order.

01

Problem and Objectives

Why fixed timers fail, the exact problem I am solving and what I aim to deliver.

02

Literature Review

What earlier researchers built, what worked, and the gap that is still open.

03

Proposed Methodology

Simulator, fuzzy controller, Genetic Algorithm tuning and how I will measure it.

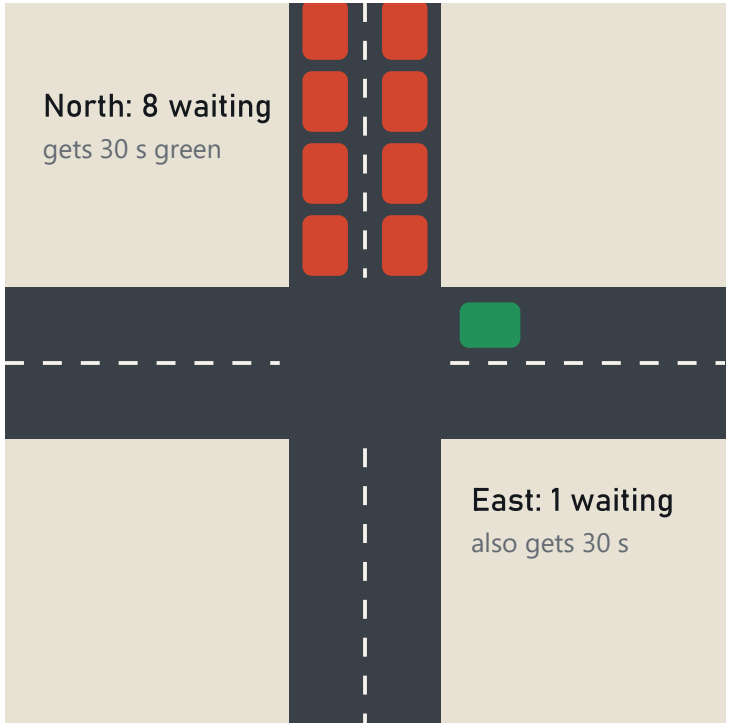
04

Plan for Review 2

Work phases, tools and the results I expect to show next time.

A fixed timer cannot see the road

Most signals give every direction the same green time, whether the road is full or empty.



Illustrative example of one signal cycle

Green time is wasted

The empty road holds a green light while the crowded road keeps waiting on red.

Traffic keeps changing

Morning peak, afternoon lull, a sudden event. One fixed plan cannot suit all of them.

The cost is real

Longer waiting means more fuel burnt while idling, more emissions and delayed emergency vehicles.

A traffic police officer does this job better because they **judge** the queue as short, medium or long. That human style of reasoning is exactly what fuzzy logic models.

Problem statement

To design a traffic signal controller for a four-way intersection that decides green time from the live queue on each road using fuzzy logic, and to tune that controller with an optimization algorithm so that the average waiting time of vehicles is as low as possible.

In scope

- One isolated four-way intersection
- Simulated traffic with random arrivals
- Fuzzy controller plus Genetic Algorithm tuning
- Comparison against a fixed-time signal

Out of scope

- Real cameras, sensors or road hardware
- A network of many connected junctions
- Pedestrian phases and turning lanes
- Deployment on a real controller

Objectives

Five things I will build and measure. Each one can be checked at Review 2.

1. **Simulate** a four-way signalised intersection in Python with random vehicle arrivals on each road.
2. **Design** a Mamdani fuzzy controller that reads queue lengths and outputs how long to extend the green.
3. **Optimize** the membership functions of that controller with a Genetic Algorithm to minimise average waiting time.
4. **Compare** three controllers on the same traffic: fixed-time, fuzzy, and GA-tuned fuzzy.
5. **Test** under light, heavy and unbalanced traffic to see where each controller wins or fails.

What has already been done

Work	Approach	Contribution	Limitation
Webster, 1958 [1]	Fixed-time formula	Gave a formula for the best cycle length and green split from average flow.	Assumes steady demand. Cannot react to sudden changes.
Pappis and Mamdani, 1977 [3]	Fuzzy controller	First fuzzy logic controller for a traffic junction. Reported lower delay than a vehicle-actuated controller.	Simple junction of two one-way roads. Rules written by hand.
Hunt et al., 1981 [4]	SCOOT adaptive system	Adjusts signal timings online from road detector data across a city network.	Needs detectors and central infrastructure. Costly to install.
Karr, 1991 [5]	GA + fuzzy	Showed that a Genetic Algorithm can tune fuzzy membership functions automatically.	Shown on general control problems, not on traffic.
Trabia et al., 1999 [6]	Two-stage fuzzy	Split the decision into two fuzzy stages for an isolated intersection with turning traffic.	Membership functions still set by expert judgement.
Niittymaki and Pursula, 2000 [7]	Fuzzy vs actuated	Simulation study showing fuzzy control can compete with traditional vehicle-actuated control.	Rule base tuned manually for the tested site.

What the literature tells me, and the gap

A

Fuzzy logic fits the problem

Queue length is naturally vague. "Long queue" has no sharp cut-off, and fuzzy sets handle that well.

B

Hand tuning is the weak point

In most fuzzy signal papers the shapes of the membership functions come from expert guesswork.

C

Optimizers can do the tuning

GA and PSO search for good parameters without needing a formula for the objective.

THE GAP I AM ADDRESSING

A clear side by side study, on one simple and reproducible simulator, of how much a fuzzy signal improves over a fixed timer, and how much more is gained when its membership functions are tuned by a Genetic Algorithm instead of by hand.

Course concepts used: Intelligent agents | Fuzzy sets and inference | Genetic Algorithm | Objective functions

System overview

The signal works as an agent: it senses the queues, reasons with fuzzy rules, then acts on the lights.

1. Traffic simulator
Four-way intersection. Vehicles arrive randomly on each road (Poisson arrivals).

→

2. Sensing
Queue on the road that has green. Longest queue among roads on red.

→

3. Fuzzy controller
Fuzzify inputs, apply rules, defuzzify with the centroid method.

→

4. Action
Extend the current green by the computed seconds, or switch phase.

↑ ↓

Optimization loop (offline)

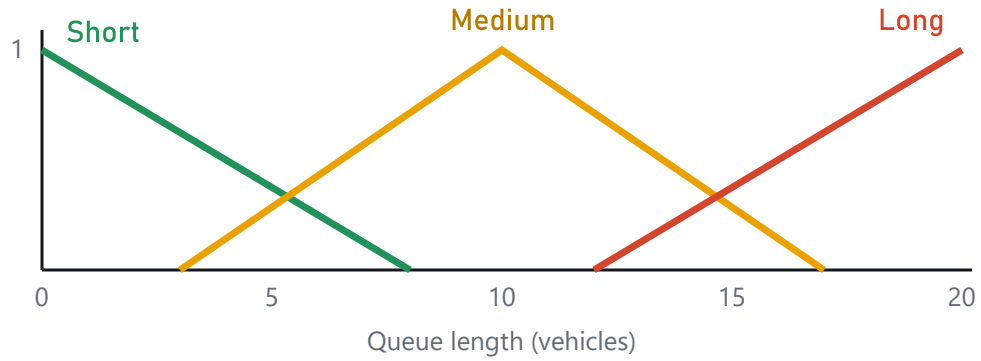
A **Genetic Algorithm** proposes new membership function shapes, runs the simulator with them, and keeps the ones that give the **lowest average waiting time**. The best set is then fixed inside the fuzzy controller.

Fuzzy controller design

Variables

Input 1: queue on the green road (vehicles)
Input 2: queue on the waiting roads (vehicles)
Output: green extension (seconds)
Each one has three fuzzy sets: Short, Medium, Long.

Membership functions (initial shapes)



Rule base (9 rules)

Cell value is the green extension.

Green road queue	Waiting: Short	Waiting: Medium	Waiting: Long
Short	Short	Short	Short
Medium	Medium	Medium	Short
Long	Long	Medium	Medium

Example: IF green road queue is **Long** AND waiting queue is **Short** THEN extension is **Long**.

Inference: Mamdani, AND as minimum.
Defuzzification: centroid, giving one crisp value in seconds.

Optimization layer: Genetic Algorithm

The rules stay the same. The GA only searches for better membership function shapes.

1. Encode
Chromosome = the corner points of all triangular membership functions.

↓

2. Evaluate
Run the simulator with each chromosome and record average waiting time.

↓

3. Select
Keep the chromosomes with the lowest waiting time (tournament selection).

↓

4. Crossover and mutate
Mix good parents and add small random changes to explore new shapes.

↻ repeat for a fixed number of generations

Fitness function

Minimise: average waiting time per vehicle

measured over one full simulation run, with the same traffic seed for every candidate so the comparison is fair.

Why a Genetic Algorithm?

The fitness comes out of a simulation, so there is no equation to differentiate. Gradient descent cannot be applied directly. GA only needs the score of each candidate.

Constraint handling

Corner points are kept in order (Short before Medium before Long) so every candidate is still a valid fuzzy set.

How I will evaluate it

Controller	How it decides green time	Role
Fixed-time	Same green for every road, every cycle	Baseline
Fuzzy (hand tuned)	Fuzzy rules with my initial membership functions	Proposed, stage 1
Fuzzy + GA	Same rules, membership functions tuned by GA	Proposed, stage 2

Traffic scenarios

Light: low arrival rate on all roads.
Heavy: high arrival rate on all roads.
Unbalanced: one busy road, three quiet roads. This is where a fixed timer should struggle most.

Metrics

- Average waiting time per vehicle (main metric)
- Average and maximum queue length
- Throughput: vehicles cleared in the run

Tools

Python | NumPy | scikit-fuzzy | Matplotlib | Custom GA code

Fair testing: all three controllers face the same random traffic, repeated over several seeds, and I report the average.

04 PLAN FOR REVIEW 2

Work plan and expected outcome

1

Build the simulator

Intersection model, arrivals, queues and the fixed-time baseline.

2

Add the fuzzy controller

Membership functions, nine rules, centroid defuzzification.

3

Tune with GA

Encode, run generations, plot the fitness curve.

4

Compare and report

Three controllers, three scenarios, graphs and conclusion.

What I expect to show

- Fuzzy control waits less than the fixed timer, most clearly in unbalanced traffic
- GA tuning gives a further drop over the hand-tuned version
- A fitness curve that improves across generations

Deliverables for Review 2

- Working Python code with a live demo run
- Comparison graphs and a results table
- Conclusion, limits found and future scope

References

[1] F. V. Webster, "Traffic Signal Settings," Road Research Technical Paper No. 39, HMSO, London, 1958.

[2] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, no. 3, pp. 338-353, 1965.

[3] C. P. Pappis and E. H. Mamdani, "A fuzzy logic controller for a traffic junction," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 7, no. 10, pp. 707-717, 1977.

[4] P. B. Hunt, D. I. Robertson, R. D. Bretherton and R. I. Winton, "SCOOT: A traffic responsive method of coordinating signals," TRRL Laboratory Report 1014, 1981.

[5] C. L. Karr, "Genetic algorithms for fuzzy controllers," *AI Expert*, vol. 6, no. 2, pp. 26-33, 1991.

[6] M. B. Trabia, M. S. Kaseko and M. Ande, "A two-stage fuzzy logic controller for traffic signals," *Transportation Research Part C*, vol. 7, no. 6, pp. 353-367, 1999.

[7] J. Niittymaki and M. Pursula, "Signal control using fuzzy logic," *Fuzzy Sets and Systems*, vol. 116, no. 1, pp. 11-22, 2000.

[8] J. H. Holland, *Adaptation in Natural and Artificial Systems*, University of Michigan Press, 1975.

Thank you



I am happy to take your questions.

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